

Maritime Chokepoint Accumulation

A Narrows Data Brief | June 2026

Outputs from the Narrows Commodity Dependency Model (CDM) and Port Commodity Index (PCI). The CDM maps annual bilateral trade values against chokepoint routing for 52 importing nations across 14 commodity classes. Base year 2022; 2023 bilateral estimates for the largest flows. Values represent annual cargo at risk per chokepoint, not simultaneous in-transit exposure.

These figures are offered as a resource for analysis and further work. Contact: fysh@narrows.io

CHOKEPOINT CLOSURE SCENARIOS

Exhibit 1 | Chokepoint Closure Scenarios

Three of nine modelled chokepoints have no maritime rerouting option. Hormuz, Bosphorus, and the Danish Straits are structural traps: when they close, trapped cargo has no sea-route alternative.

Chokepoint	Annual cargo at risk	Trapped (no reroute)	Maritime reroute?	Detour penalty	Largest exposure
Strait of Hormuz	\$630B	\$630B 100%	No	None	China crude \$167B
Bosphorus	\$125B	\$125B 100%	No	None	Germany crude \$22B
Danish Straits	\$27B	\$27B 100%	No	None	Poland crude \$11B
Strait of Malacca	\$604B	n/a	Yes	+5 days	China crude \$202B
Lombok Strait	\$277B	n/a	Yes	+3 days	China iron ore \$121B
Strait of Gibraltar	\$199B	n/a	Yes	+20 to 38d	Germany crude \$29B
Suez Canal	\$153B	n/a	Yes	+16 days	Italy crude \$15B
Bab el-Mandeb	\$153B	n/a	Yes	+16 days	Italy crude \$15B
Panama Canal	\$62B	n/a	Yes	+38 days	Korea LNG \$10B

Suez and Bab el-Mandeb values are identical: the same Red Sea flows counted at each chokepoint transited. Gibraltar detour range reflects per-corridor bypass distances: Atlantic-to-Med routes require Cape of Good Hope plus Suez from the east (+20 to 38 days depending on origin corridor).

LARGEST TRAPPED BILATERAL EXPOSURES

Exhibit 2 | Largest Trapped Bilateral Exposures

The 20 highest-value bilateral flows with no maritime reroute option. Italy crude appears at both Hormuz and Bosphorus: two separate import corridors, both unbypassable.

Importing country	Commodity	Chokepoint	Dependency	Annual value
China	Crude petroleum	Hormuz	46%	\$167.5B
India	Crude petroleum	Hormuz	59%	\$69.6B
Japan	Crude petroleum	Hormuz	83%	\$67.1B
Rep. of Korea	Crude petroleum	Hormuz	73%	\$62.9B
Thailand	Crude petroleum	Hormuz	71%	\$21.9B
Germany	Crude petroleum	Bosphorus	36%	\$21.7B
Singapore	Crude petroleum	Hormuz	73%	\$20.0B
Taiwan	Crude petroleum	Hormuz	70%	\$18.8B
Netherlands	Crude petroleum	Bosphorus	31%	\$17.0B
India	Crude petroleum	Bosphorus	14%	\$16.8B
Italy	Crude petroleum	Hormuz	40%	\$15.4B
China	LNG	Hormuz	24%	\$11.7B

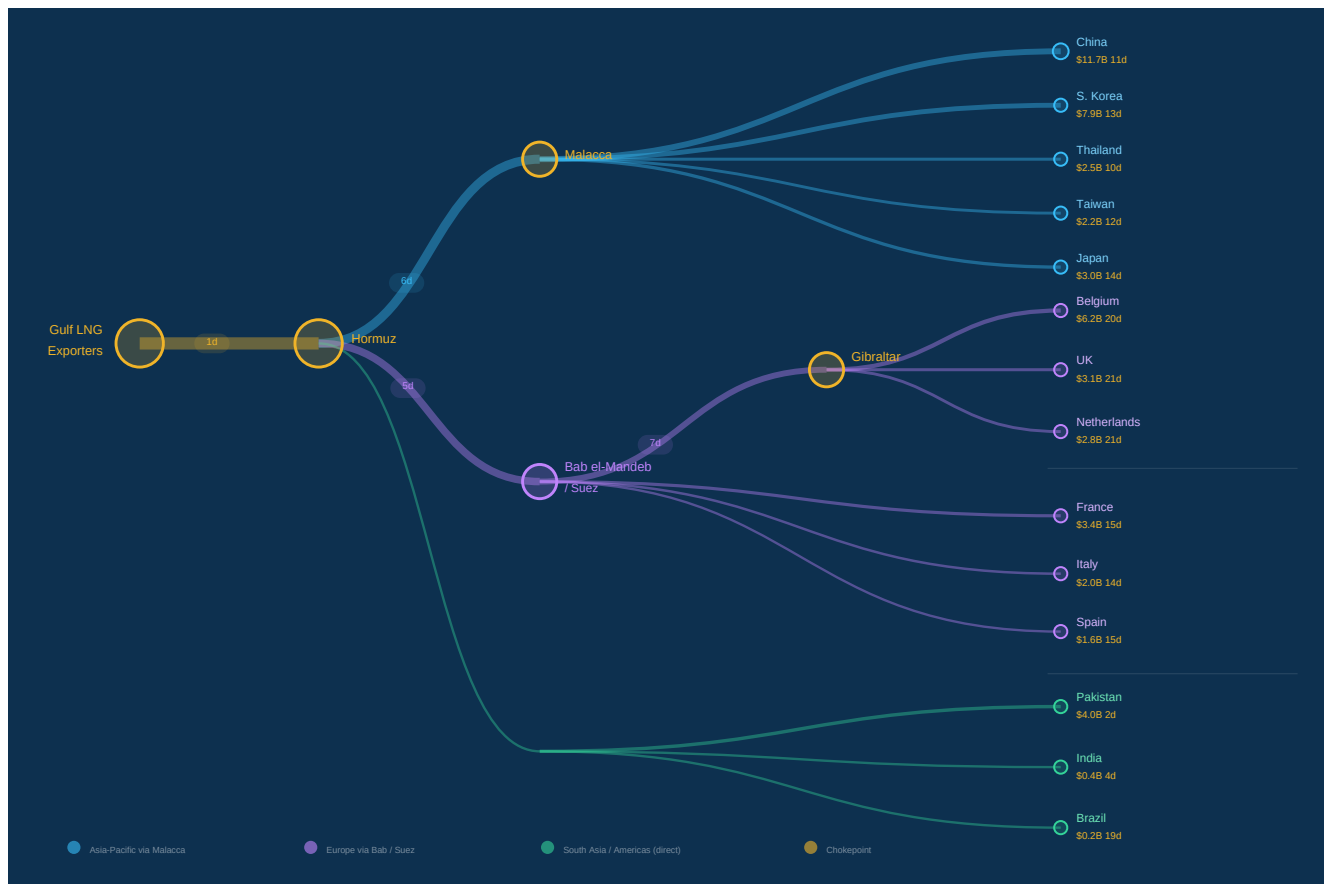
Importing country	Commodity	Chokepoint	Dependency	Annual value
Netherlands	Crude petroleum	Hormuz	21%	\$11.5B
Poland	Crude petroleum	Danish Str.	95%	\$11.2B
Italy	Crude petroleum	Bosphorus	26%	\$10.0B
Spain	Crude petroleum	Hormuz	27%	\$9.7B
Greece	Crude petroleum	Hormuz	55%	\$8.5B
Turkey	Crude petroleum	Hormuz	53%	\$8.5B
Rep. of Korea	LNG	Hormuz	20%	\$7.9B
France	Crude petroleum	Bosphorus	18%	\$6.4B

Dependency = share of that commodity's total annual imports exposed to this chokepoint. All entries sourced from UN Comtrade bilateral pair data.

QATAR LNG SEQUENTIAL CHOKEPOINT CHAIN

Exhibit 3 | The Qatar LNG Sequential Chokepoint Chain

Every cargo leaving Ras Laffan must pass through Hormuz. No maritime alternative exists. Onward routing splits: Asia-Pacific via Malacca; Europe via Bab el-Mandeb and Suez. Atlantic-facing terminals (Belgium, UK, Netherlands) add Gibraltar as a fourth sequential chokepoint. Each stage is independent: disruption at any point breaks the entire chain.



Hormuz is the sole mandatory exit for all Gulf LNG. No pipeline bypass and no alternative maritime route exists for laden LNG tankers. Italy, France, and Greece receive Qatari LNG via Suez into Mediterranean terminals (3 sequential chokepoints). Belgium, the UK, and the Netherlands receive via Gibraltar into Atlantic terminals (4 sequential chokepoints). These dependencies are structural, not probabilistic.

Exhibit 3a | European LNG Exposure by Supply Route

Belgium, UK, and Netherlands import LNG exclusively at Atlantic-facing terminals. Their entire Qatar supply transits all four chokepoints in sequence. Spain, Italy, and France have mixed Med and Atlantic terminal capacity; their Gibraltar exposure reflects a different, Atlantic-origin supply chain.

Importer	Annual LNG value	Hormuz (trapped)	Bab el-Mandeb	Suez	Gibraltar	Notes
Belgium	\$6.2B	56%	56%	56%	57%	Atlantic terminal; all 4 chokepoints on same Qatar chain
United Kingdom	\$3.1B	26%	26%	26%	26%	Atlantic terminal; all 4 chokepoints on same Qatar chain
Netherlands	\$2.8B	12%	12%	12%	16%	Atlantic terminal; all 4 chokepoints on same Qatar chain
France	\$3.4B	10%	11%	11%	20%	Hormuz/BaM/Suez = Qatar via Med (Fos); Gib excess = Montoir Atlantic
Italy	\$2.0B	22%	24%	24%	61%	Hormuz/BaM/Suez = Qatar via Med; Gib excess = Atlantic-origin LNG
Spain	\$1.6B	17%	17%	17%	69%	Hormuz/BaM/Suez = Qatar to Med; Gib 69% includes heavy Atlantic supply
Pakistan	\$4.0B	90%	n/a	n/a	n/a	Trapped; Hormuz only; no reroute
Lithuania	\$4.3B	n/a	n/a	n/a	100%	Danish Straits 100% trapped; see Exhibit 4

Hormuz dependency is trapped (no maritime reroute) for all entries. Gibraltar dependency for Spain and Italy exceeds their Suez dependency because Atlantic-origin LNG (US, Nigeria, Norway) transits Gibraltar but not the Red Sea route.

OTHER HIGH-CONCENTRATION EXPOSURES

Exhibit 4 | Other High-Concentration Exposures

Iron ore, coal, and Pacific LNG supply chains with material single-chokepoint concentration.

Importer	Commodity	Chokepoint	Dep.	Annual value	Reroute?	Note
China	Iron ores	Lombok	96%	\$120.8B	Yes +3d	Australian Capesize fleet; Cape route available
Japan	Iron ores	Lombok	99%	\$13.6B	Yes +3d	Near-total Lombok dependence
Rep. of Korea	Iron ores	Lombok	98%	\$8.8B	Yes +3d	
Lithuania	LNG	Danish Straits	100%	\$4.3B	No	Klaipeda: sole Baltic LNG terminal; fully trapped
Pakistan	LNG	Hormuz	90%	\$4.0B	No	100% Gulf-sourced; no maritime bypass
Bangladesh	LNG	Hormuz	82%	\$2.0B	No	
Rep. of Korea	LNG	Panama	25%	\$9.7B	Yes +38d	US Atlantic LNG; Cape Horn or COGH reroute
Japan	LNG	Panama	15%	\$7.4B	Yes +38d	US Atlantic LNG supply
Viet Nam	Coal	Luzon	65%	\$4.3B	Yes +2d	South China Sea corridor

UPSTREAM PORT CONCENTRATION

Exhibit 5 | Upstream Port Concentration

The Port Commodity Index maps 146 terminals across 10 maritime basins. These are the upstream single points of failure behind the chokepoint model.

Commodity	Port	Country	Basin	Throughput (Mt/yr)	Share of indexed total
LNG	Ras Laffan Industrial City	Qatar	Gulf	77	43%
LNG	Sabine Pass LNG	USA	Atlantic	45	25%
LNG	Bintulu MLNG	Malaysia	SCS	30	17%
Iron ore	Port Hedland	Australia	Indian	450	55%
Iron ore	Port of Dampier	Australia	Indian	150	18%
Iron ore	Porto de Tubarao	Brazil	Atlantic	120	15%

Commodity	Port	Country	Basin	Throughput (Mt/yr)	Share of indexed total
Crude	Ras Tanura / Juaymah	Saudi Ar.	Gulf	324	26%
Crude	Basra Oil Terminal	Iraq	Gulf	180	14%
Crude	Jebel Dhanna / Ruwais	UAE	Gulf	145	12%
Coal	Port of Newcastle	Australia	Pacific	160	43%
Coal	Port of Gladstone	Australia	Pacific	67	18%

Ras Laffan sustained damage in the 2026 Hormuz conflict (Wood Mackenzie, March 2026). Shares reflect indexed ports only; total global throughput is higher.

REROUTING COST PREMIUMS

Exhibit 6 | Rerouting Cost Premiums

When rerouting is available, it is not cost-neutral. Extended transit raises freight costs, stretches war risk coverage periods, and creates accumulation at alternative corridors.

Closure	Primary reroute	Extra transit (CDM avg)	Additional cost/voyage	Notes
Suez / Bab el-Mandeb	Cape of Good Hope	+15.6 days	\$800k to \$1.9M	VLCC crude: 15.6d x \$120k/d = \$1.87M. Aframax refined: \$780k. LNG carrier: \$1.53M
Panama Canal	Cape Horn	+38.1 days	\$2.1M to \$3.7M	LNG carrier: 38.1d x \$98k/d = \$3.73M. Container: 38.1d x \$55k/d = \$2.10M
Malacca Strait	Lombok / Sunda	+2.6 days	\$175k to \$310k	Capesize bulk: 2.6d x \$67k/d = \$174k. VLCC crude: 2.6d x \$120k/d = \$312k
Gibraltar	COGH then Suez from east	+20 to 38 days	\$2.4M to \$4.6M	Atlantic-to-Med routes. VLCC via West Africa: 32.9d x \$120k/d = \$3.95M. LNG exiting Med: 38d x \$98k/d = \$3.72M
Hormuz	None maritime	No reroute	n/a	No bypass for laden tankers. Pipeline alternatives limited and pre-committed
Bosphorus	None maritime	No reroute	n/a	Black Sea is semi-enclosed. No maritime bypass for outbound cargo
Danish Straits	None maritime	No reroute	n/a	Baltic is semi-enclosed. Poland crude and Lithuania LNG fully trapped

Costs derived from Narrows CDM vessel rate model (Baltic Exchange baseline, May 2026): VLCC \$120k/day (bunker + charter); LNG Carrier \$98k/day; Capesize \$67k/day; Aframax \$50k/day. Formula: additional cost = (bunkerPerDay + charterPerDay) x avg_detour_days. Detour days from CDM corridor model. Gibraltar costs apply to Atlantic-to-Mediterranean routes only (per-corridor Fix-33 methodology).